The electric dipole moment (EDM) of fundamental particles reflects the distribution of electric charge within that particle and is commonly measured to search for physics beyond the standard model. Finding EDM values not predicted by the standard model could provide insight into some of the biggest mysteries in physics, such as matter-antimatter asymmetry. The Odom group proposes using atoms trapped inside inert crystals (doped crystals) to make EDM measurements with much higher sensitivities than previous experiments. Before this, however, we must define and optimize the best way to grow these doped crystals. My part in this project will be to develop and characterize a rubidium (Rb) atomic oven to consistently supply a dopant density of 10^{13} atoms per cubic millimeter during testing.

Atomic ovens are very common. They heat up and then direct dopants, like Rb, into test chambers to perform many different experiments; however, our oven will have to deal with circumstances not nearly as common or previously characterized. Since the first step of our main experiment is finding the best way to grow large, doped crystals, we will vary multiple parameters, the most impactful of which for the oven will be the pressure of our chamber. When growing our crystal, the inert gas argon (Ar) will pass through our oven before filling up the growth chamber, where it will form a crystal on a growth plate. During this process, we may use pressures ranging from 10^{-6} torr to one torr. However, most Rb ovens are designed to operate at one pressure, usually between 10^{-10} torr to 10^{-5} torr [3][4]. Operating at higher pressures is not necessarily a problem, though, since this just requires us to run the oven at a hotter temperature. This increases the vapor pressure/amount of Rb in the oven to maintain the ratio of Ar to Rb associated with our desired dopant density. People have done this in similar crystal growth experiments to operate ovens at pressures above one torr [5]; however, they still maintained a constant pressure, so the effect of actively varying the pressure in an atomic oven is still poorly characterized.

Another problem occurs when the Rb leaves the oven and travels to the growth plate. At high enough pressures, inert gasses have been shown to cause an exponential decrease in atomic flux (number of atoms passing a unit area every second) with both an increase in distance and pressure [1]. We should be able to avoid this problem altogether, however, by keeping both Rb and Ar in molecular flow, which occurs when atoms rarely collide/interact with each other before hitting a wall. We can also combat this by increasing the overall flux and collimating the Rb beam from the oven. A method previously used to increase flux and collimate the beam leaving an atomic oven is using an array of microcapillaries (small tubes of inner diameter around 100 μm). These microcapillaries have been shown to create collimated atom beams with divergence angles (angle our beam deviates from a straight path) of around only 1.2 degrees [2]. Microcapillaries have also been shown to produce a high atomic flux of around $7x10^{11}$ atoms/sec per microcapillary [3]. So, with an array of approximately 150 similar microcapillaries, we could achieve an atomic flux out of our oven on the order of 10^{14} atoms/sec, which should be sufficient for achieving our desired dopant density. Additionally, due to these capillaries' very small inner diameter, the Rb and Ar leaving the oven will almost certainly be in molecular flow.

The oven itself will have three main parts. A side arm containing the Rb, a main chamber with windows to conduct saturated absorption spectroscopy (SAS)–a technique that uses lasers to tell us the amount of Rb in our chamber–and a nozzle with a microcapillary array connecting the oven to the growth chamber. Each section will also have its own heating elements for more refined control over oven temperatures. After building this oven, I will characterize how changing the chamber pressure and oven temperature affects atomic flux and beam divergence.

Then, I will figure out how to consistently maintain our desired dopant density–directly related to flux and divergence–for all pressures we may operate at.

To start characterizing the oven, I will connect it to a test chamber with additional sets of windows to allow for SAS at the nozzle of the oven and the growth plate location. This, along with the windows in the oven, will allow us to see how much Rb is at each point in our chamber. SAS has also been used to characterize atomic beams by finding atomic flux and defining beam divergence [6]. My first task will be to make the Rb beam is as collimated as possible by varying the different temperatures of each part of the oven; it is common to have the nozzle of ovens hotter than the main chamber of the oven [2][3][4], so I will start with this. I aim to match the previously mentioned beam divergence of 1.2 degrees at the growth plate and atomic flux of order 10^{14} leaving the nozzle at a low chamber pressure (around 10^{-7} torr). I will also use this phase of my project to ensure my design can reach these criteria; if divergence is too large and flux too low, I can modify the microcapillary array by using thinner and longer capillaries or increase the temperature of the nozzle, both methods should decrease divergence and increase flux. These changes will be easy due to the modular three-part design. Once I meet these specifications, I will maintain temperatures and increase pressure while continuing to measure beam divergence and atomic flux. This data will give me a better understanding of how sensitive my oven is to chamber pressure. Next, I will maintain pressure while varying oven temperature and continue to measure flux and divergence. With these two data sets, I can see how flux and divergence are offset from our target at a certain pressure, and then I can look at temperature data to see how much I need to change temperature to undo this offset. This method may not work exactly, but it will provide a starting point at which, if still off of the desired divergence and flux, I can change temperature beyond what the temperature data predicts. Doing this several times will provide insight into how ovens operate at varying pressures and allow me to identify how to maintain our desired dopant density for all the pressures we work with.

I will also have lots of experience and mentors I can rely on during this project. First, last spring and summer, I worked in Professor Kovachy's lab designing a vapor cell, which, like an oven, reaches high temperatures to get atoms, like Rb, into a gaseous phase. During this project, I spent several hours designing the cell in Solidworks, machining prototype parts, and testing different heating elements. All of these skills will be helpful in my oven project. Additionally, the Ph.D. student I worked with during that project has experience with ovens and expressed her willingness to help me with any questions that arise during this new project. I now have been working in Professor Odom's lab since the fall quarter, helping design the main crystal growth chamber and designing the oven. I attend weekly meetings, both with the entire lab and with a Ph.D. student, allowing me to get an in-depth understanding of the overall goals of the lab and my project. Additionally, with the completion of Physics 339-1/339-2 (Quantum Mechanics) by summer, I will have a much better general understanding of the physics our lab will work with, and with the completion of Physics 332-0 (Statistical Mechanics), I will have a much better understanding on how to do calculations to understand how Rb and Ar behave in both our oven and crystal growth chamber.

During this project, I will learn many skills that will help me in my future career in physics. This project will be led by me, which will teach me how to conduct my own research and experiments, as well as develop my decision-making skills. This project could also be used in part of an honors thesis I write. Additionally, atomic physics is of interest to me and a field of physics that I may continue to work in in the future, so having a good understanding of how atomic ovens work (which are very common in the field) will be very helpful.

References

[1] Egorov, D., Lahaye, T., Schöllkopf, W., Friedrich, B., & Doyle, J. M. (2002). Buffer-gas cooling of atomic and molecular beams. In Physical Review A (Vol. 66, Issue 4). American Physical Society (APS). https://doi.org/10.1103/physreva.66.043401

[2] Senaratne, R., Rajagopal, S. V., Geiger, Z. A., Fujiwara, K. M., Lebedev, V., & Weld, D. M. (2015). Effusive atomic oven nozzle design using an aligned microcapillary array. In Review of Scientific Instruments (Vol. 86, Issue 2). AIP Publishing. https://doi.org/10.1063/1.4907401

[3]Lin, Y.-J., Perry, A. R., Compton, R. L., Spielman, I. B., & Porto, J. V. (2009). Rapid production of ^{87}Rb Bose-Einstein condensates in a combined magnetic and optical potential. In Physical Review A (Vol. 79, Issue 6). American Physical Society (APS). https://doi.org/10.1103/physreva.79.063631

[4] Brown-Hayes, M., Wei, Q., Kim, W.-J., & Onofrio, R. (2007). Development of an apparatus for cooling 6Li-87Rb Fermi-Bose mixtures in a light-assisted magnetic trap. In Laser Physics (Vol. 17, Issue 4, pp. 514–522). Pleiades Publishing Ltd. https://doi.org/10.1134/s1054660x07040330

[5] Kanagin, A. N., Regmi, S. K., Pathak, P., & Weinstein, J. D. (2013). Optical pumping of rubidium atoms frozen in solid argon. In Physical Review A (Vol. 88, Issue 6). American Physical Society (APS). https://doi.org/10.1103/physreva.88.063404

[6] Sabulsky, D. O. (2014). *A Cesium-133 effusive oven for ultracold atomic experiments* (Doctoral dissertation, Ph. D. thesis, the university of chicago, Chicago).

Covid Back Up Plan

A good part of my project involves data analysis, so if I happen to get covid, most of this work can be done asynchronously. Data measurements can be sent to me from the lab and I can join any group meetings or advisor meetings over Zoom. Additionally, since I am starting near the earlier end of the summer, if the need arises, I could continue this project and work with the lab for an additional week, or whatever is needed, past the planned end date.